

MATH3026/4022 Lecture 7

Model Identification and Fitting I

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19th February 2026

- ▶ General approach to model identification
- ▶ Use and interpretation of sample plots
- ▶ Parameter estimation for the $AR(1)$ process:
 - ▶ Method of moments estimation
 - ▶ Maximum likelihood estimation

Model Identification: Introduction

In Lecture 6 we explored sample statistics and their properties for time series, where we have an observed sequential sample of time series data

$$\mathbf{x} = (x_1, \dots, x_n)^\top.$$

from some time series.

A natural question is then ‘which type of model best fits our observed data?’

In this Lecture, we’ll explore this in more detail, including how sample statistics might be used to guide our choice of model.

Model Identification: $ARMA(p, q)$

We describe a general approach for identifying the type of $ARMA(p, q)$ model that might best fit our observed data.

We note that much of the approach is useful for fitting any type of time series model.

Unsurprisingly, and as with any statistical inference approach, we begin with basic summaries of the set of observed data $\mathbf{x} = (x_1, \dots, x_n)^\top$.

Our approach will be summarised as a series of steps in the next few slides.

Model Identification: $ARMA(p, q)$

1. Produce a time plot of time against the observed values $x_1, \dots, x_n$.

Examining this plot, we ask the questions:

- (i) Does the time series look stationary?
- (ii) Does the time series have seasonal components?

As a result, we might consider transforming the time series (e.g. by using differencing or perhaps a filter - covered later in the course) to make it stationary.

Alternatively, methods for non-stationary time series may be used (covered later in this course).

Model Identification: $ARMA(p, q)$

2. Calculate the sample autocorrelations for a number of lags (e.g. up to 25-30) and then produce a plot of r_h against the lag, h .

This type of plot is known as a **correlogram** and is very useful when identifying models to fit.

For example, if we see that all of the values, r_h , lie close to zero then we might conclude that successive observations are uncorrelated and that the process is white noise.

Alternatively, if the values r_h appear to drop to zero after lag q , then we might conclude that an $MA(q) = ARMA(0, q)$ model is appropriate.

Model Identification: $ARMA(p, q)$

3. Calculate the sample partial autocorrelations, $\hat{\beta}_{m,m}$ for a number of lags (e.g. up to 25-30) and then produce a plot of $\hat{\beta}_{m,m}$ against the lag, m .

As with the correlogram, this plot should be compared with expected theoretical forms for different time series models.

For example, we have seen that if a process is purely an $AR(p)$ process (i.e. an $ARMA(p, 0)$ process), then the sample PACF should 'cut off' to zero after lag p .

If we don't see this behaviour in a plot of the sample PACF then we might assume that the $ARMA$ model includes a moving average component.

Model Identification: $ARMA(p, q)$

4. Once we've chosen a model to fit to the data, we estimate the parameters of the model.

Approaches to do this will be covered later in this lecture.

5. As with any fitted model, once parameter estimates are obtained in Step 4, we should check that the model is good fit to the data (and consider alternatives if not).

Model checking will be covered in later lectures.

Sample Plots for Common Time Series

We now simulate some 'observed' data from common time series models and explore the plots outlined in Steps 1-3.

The command `arima.sim` in R is used to simulate data.

For example, to simulate 200 observations from an $ARMA(1,1)$ process with $\phi_1 = 0.5$ and $\theta_1 = 0.6$, we would write

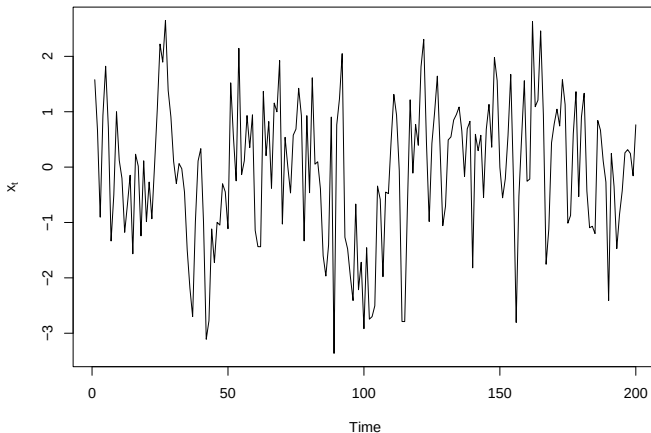
```
x<-arima.sim(model=list(ar=0.5,ma=0.6),200).
```

The commands `acf(x)` and `pacf(x)` can then be used to produce plots of the sample autocorrelation against the lag and the sample partial autocorrelation against the lag, respectively.

Sample plots: $AR(1)$ process, $\phi_1 = 0.6$

We begin by simulating 200 observations from an $AR(1)$ process where $\phi_1 = 0.6$. A time plot of the observed data is shown below.

Time plot for simulated $AR(1)$ data, $\phi_1 = 0.6$



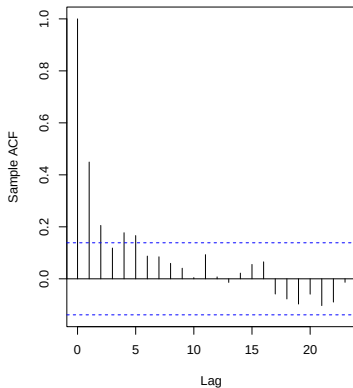
Sample plots: $AR(1)$ process, $\phi_1 = 0.6$

As expected, the observed time series data appear to be stationary with mean zero and there is no obvious seasonality.

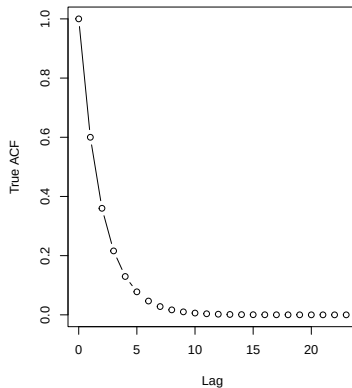
Next we consider a plot of the sample ACF against the lag and a plot of the theoretical ACF against the lag for the set of 200 simulated observations from an $AR(1)$ model where $\phi_1 = 0.6$.

Sample plots: $AR(1)$ process, $\phi_1 = 0.6$

Sample ACF vs. Lag
for simulated $AR(1)$ data



True ACF vs. lag: $AR(1)$ process, $\phi_1 = 0.6$



Sample plots: $AR(1)$ process, $\phi_1 = 0.6$

Note that the horizontal dashed blue lines on the left hand plot are at $\pm \frac{2}{\sqrt{n}}$.

In theory, we know that the ACF for the $AR(1)$ process at lag h is $\phi_1^{|h|}$, so where $\phi_1 \in (0, 1)$ we expect the ACF to decay exponentially.

Indeed, that is what we observe in the sample ACF plot (correlogram).

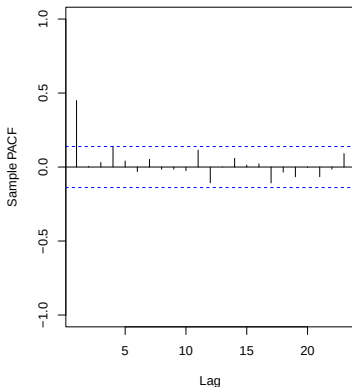
The sample ACF values decrease steadily towards zero as the lag increases.

If we did not already know that our data were from an $AR(1)$ process, we'd look at this plot and infer that it's likely that the process from which the data arise includes an $AR(p)$ component where p is small.

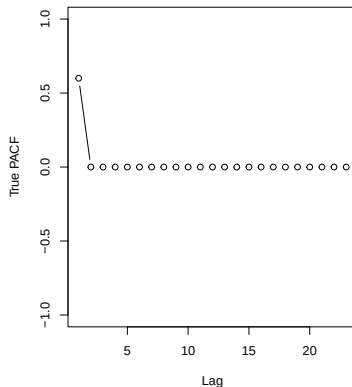
Sample plots: $AR(1)$ process, $\phi_1 = 0.6$

Now let's consider the sample partial autocorrelation function

Sample PACF vs. Lag
for simulated $AR(1)$ data



True PACF vs. lag: $AR(1)$ process, $\phi_1 = 0.6$



Sample plots: $AR(1)$ process, $\phi_1 = 0.6$

We observe that the sample PACF rapidly decreases to values close to zero after lag 1.

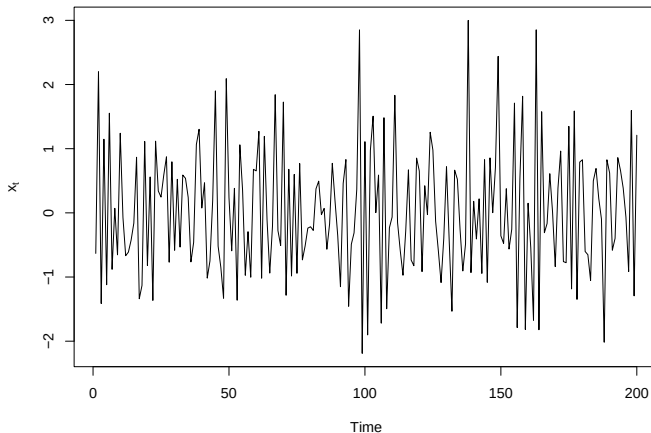
Comparing the sample PACF plot to the theoretical PACF plot, we see that this suggests that the data have arisen from a $AR(1)$ process.

Note that the correlogram caused to infer that the data were from an $AR(p)$ process but the sample PACF plot confirmed that the process is of order 1.

Sample plots: $MA(1)$ process, $\theta_1 = -0.5$

Now we simulate 200 observations from an $MA(1)$ process where $\theta_1 = -0.5$. A time plot of the observed data is shown below.

Time plot for simulated $MA(1)$ data, $\theta_1 = -0.5$



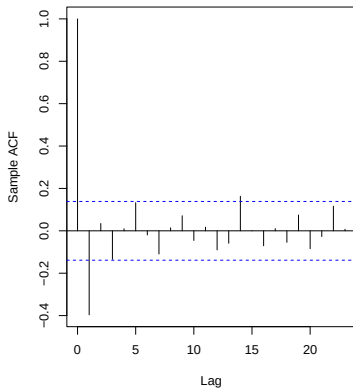
Sample plots: $MA(1)$ process, $\theta_1 = -0.5$

As we should expect, the observed time series data appear to be stationary with mean zero and there is no obvious seasonality.

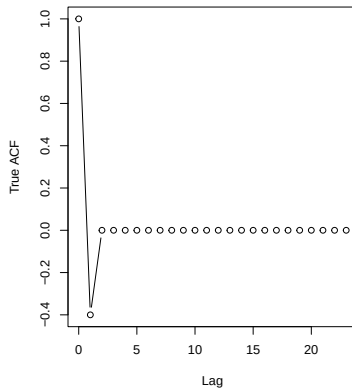
Next we consider a plot of the sample ACF against the lag and a plot of the theoretical ACF against the lag for an $MA(1)$ process where $\theta_1 = -0.5$.

Sample plots: $MA(1)$ process, $\theta_1 = -0.5$

Sample ACF vs. Lag
for simulated $MA(1)$ data



True ACF vs. lag: $MA(1)$ process, $\theta_1 = -0.5$



Sample plots: $MA(1)$ process, $\theta_1 = -0.5$

In theory, we know that the ACF for the $MA(q)$ process 'cuts off' after lag q .

Here, we see that the sample ACF appears to drop to zero (values close to zero) after lag 1.

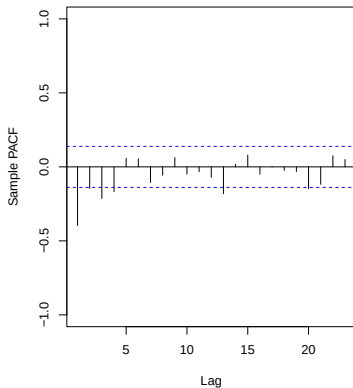
We might begin to think that the observed data originate from an $MA(1)$ process.

Moreover, the negative sample autocorrelation value at lag 1 suggests that the coefficient in the $MA(1)$ process lies in the set $(-1, 0)$.

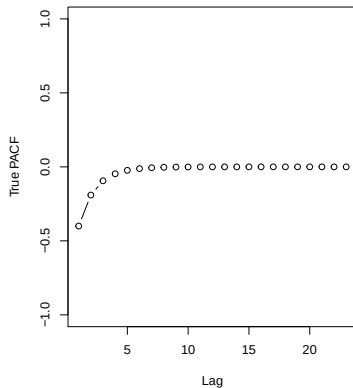
Now let's consider the sample partial autocorrelation function...

Sample plots: $MA(1)$ process, $\theta_1 = -0.5$

Sample PACF vs. Lag
for simulated $MA(1)$ data



True PACF vs. lag: $MA(1)$ process, $\theta_1 = -0.5$



Sample plots: $MA(1)$ process, $\theta_1 = -0.5$

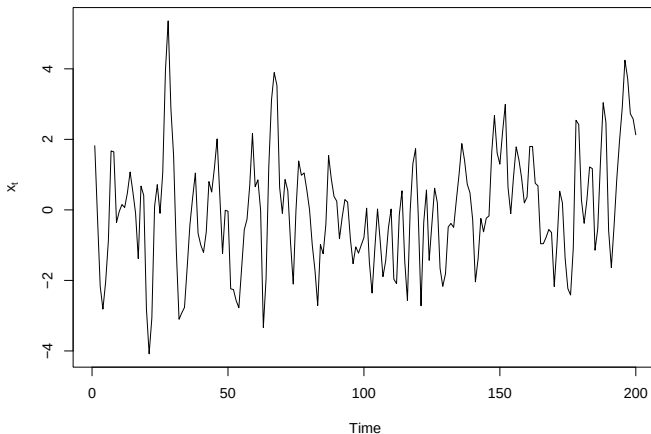
We observe that the sample PACF dampens towards zero fairly rapidly.

This is indicative of an MA process, in contrast to that for the $AR(1)$ process, where the PACF cuts to zero after lag p .

Sample plots: $ARMA(1,1)$ process, $\phi_1 = 0.5, \theta_1 = 0.7$

Now we simulate 200 observations from an $ARMA(1,1)$ process where $\phi_1 = 0.5$ and $\theta_1 = 0.7$. A time plot of the observed data is shown below.

Time plot for simulated $ARMA(1,1)$ data, $\phi_1 = 0.5, \theta_1 = 0.7$



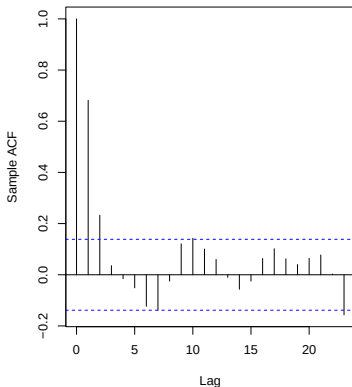
Sample plots: $ARMA(1, 1)$ process, $\phi_1 = 0.5, \theta_1 = 0.7$

As expected the observed time series data appear to be stationary with mean zero and there is no obvious seasonality.

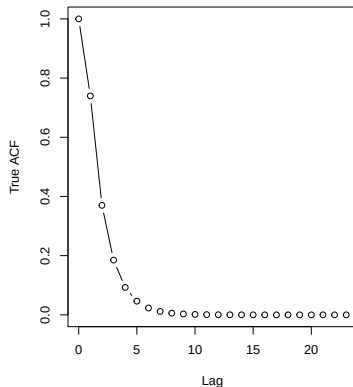
Next we consider a plot of the sample ACF against the lag and a plot of the theoretical ACF against the lag.

Sample plots: $ARMA(1,1)$ process, $\phi_1 = 0.5, \theta_1 = 0.7$

Sample ACF vs. Lag
for simulated ARMA(1,1) data



True ACF vs. lag: ARMA(1,1), $\phi_1 = 0.5, \theta_1 = 0.7$



Sample plots: $ARMA(1, 1)$ process, $\phi_1 = 0.5, \theta_1 = 0.7$

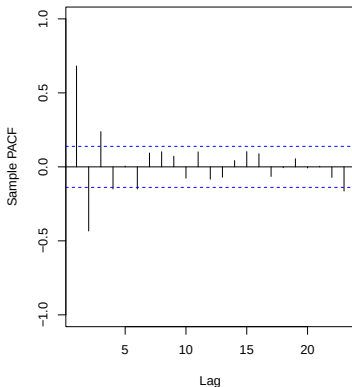
Since the process is (weakly) stationary, we see that the sample ACF decays to zero as the lag increases, which we should expect.

The presence of both an AR and an MA component means that we do not observe the sample ACF 'cutting off' to zero after a particular lag.

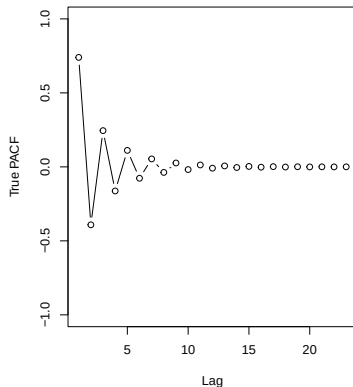
Next we consider a plot of the sample PACF against the lag and a plot of the theoretical PACF against the lag.

Sample plots: $ARMA(1,1)$ process, $\phi_1 = 0.5, \theta_1 = 0.7$

Sample PACF vs. Lag
for simulated ARMA(1,1) data



True PACF vs. lag: ARMA(1,1), $\phi_1 = 0.5, \theta_1 = 0.7$



Sample plots: $ARMA(1, 1)$ process, $\phi_1 = 0.5, \theta_1 = 0.7$

The sample PACF decays to zero as the lag increases and matches the profile of the true PACF, which we should expect.

The presence of both an AR and an MA component means that we do not observe the sample PACF 'cutting off' to zero after a particular lag.

Combining information from the sample ACF and sample PACF plots would lead us to believe that a model fitted to these data should contain both an AR and an MA component.

Model Fitting

Having observed various properties of sample statistics and plots, suppose now that we wish to fit a model of a particular type to an observed set of data.

We know that our chosen model type will include unknown parameters (e.g. ϕ_1 in an $AR(1)$ model).

Now, we consider how such parameters might be estimated using our observed set of time series data.

We'll cover two approaches to estimation:

- (i) Method of moments estimation;
- (ii) Maximum likelihood estimation.

To begin with, we'll consider estimation for the $AR(1)$ process before considering that for other models (in Lecture 8).

Estimation: $AR(1)$ process

Suppose that the model for the $AR(1)$ process is written

$$X_t - \mu = \phi_1(X_{t-1} - \mu) + Z_t$$

where Z_t is a white noise term with mean zero and variance σ_z^2 .

Note here that X_t has mean μ but we consider a model for $X_t - \mu$ which has mean zero.

The parameters that we would like to estimate are μ , ϕ_1 and σ_z^2 .

We assume that we have observed the set of time series data $\mathbf{x} = (x_1, \dots, x_n)^\top$ and that we'd like to fit an $AR(1)$ model using these data.

Here, fitting the model amounts to estimating the parameters μ , ϕ_1 and σ_z^2 using the observed data $\mathbf{x}$.

$AR(1)$: Method of moments

We recall that, when performing method of moments of estimation, we replace population moments with sample moments and solve resulting equations to obtain estimates for parameter(s) of interest.

For an $AR(1)$ process we have seen that

$$\rho(1) = \phi_1.$$

Hence, replacing $\rho(1)$ with its sample counterpart, r_1 , we obtain

$$\tilde{\phi}_1 = r_1$$

as a method of moments estimate for ϕ_1 .

AR(1): Method of moments

Furthermore, we have seen that

$$\text{Var}(X_t) = \gamma(0) = \frac{\sigma_z^2}{1 - \phi_1^2}.$$

Rearranging the above and replacing $\gamma(0)$ and ϕ_1 by their method of moments estimates (c_0 and r_1 , respectively), we obtain the following estimate for σ_z^2

$$\tilde{\sigma}_z^2 = c_0(1 - r_1^2).$$

Finally, μ may be estimated by the sample mean. That is

$$\tilde{\mu} = \bar{\mathbf{x}}.$$

Some Comments

Some comments on the method of moments approach to estimation:

1. Generally, we make no distributional assumptions about X_t .
2. Estimators are not invariant under re-parameterisation.
3. Typically, estimators are consistent.

That is, if $\tilde{\theta}_n$ is an estimator for θ , using a sample of size n , then

$$\lim_{n \rightarrow \infty} \mathbb{E}(\tilde{\theta}_n) = \theta$$

where expectation is taken with respect to the sampling distribution of the estimator $\tilde{\theta}_n$.

AR(1) Process: Maximum Likelihood Estimation

We know that maximum likelihood estimation is a parametric form of estimation.

That is, we must assume a probability distribution for our data (with parameters of interest) and then use a likelihood function to make inference about the parameters of interest.

For time series models, a common distributional assumption is to assume that the white noise terms $Z_1, Z_2, \dots$ are independent normal random variables with mean zero and variance σ_z^2 .

With this assumption, our AR(1) model is written

$$X_t - \mu = \phi_1 (X_{t-1} - \mu) + Z_t$$

with $Z_t \sim \mathcal{N}(0, \sigma_z^2)$ and Z_i, Z_j independent where $i \neq j$.

AR(1) Process: Maximum Likelihood Estimation

We see that X_t depends only on X_{t-1} (and not on X_{t-2}, X_{t-3} etc.).

This is known as the **Markov Property**.

We know that X_t can be written in the form

$$X_t = \mu + \sum_{j=0}^{\infty} \phi_1^j Z_{t-j} \quad (7.1)$$

Hence, we see that X_t is a linear combination of the normal random variables $Z_t, Z_{t-1}, Z_{t-2}, \dots$ and this implies that X_t is normally distributed (and so is X_{t-1}).

Hence, X_t and X_{t-1} have a bivariate normal distribution and the conditional distribution of X_t , given X_{t-1} , is

$$X_t \mid X_{t-1} = x_{t-1} \sim \mathcal{N}(\mu + \phi_1(x_{t-1} - \mu), \sigma_z^2).$$

$AR(1)$ Process: Bivariate Normal Distribution

AR(1) Process: Maximum Likelihood Estimation

To construct the likelihood function, we define $\boldsymbol{\theta} = (\mu, \phi_1, \sigma_z^2)^\top$ to be our vector of unknown parameters (for convenience).

The likelihood function is simply the joint density of $\mathbf{x}$, but considered to be a function of $\boldsymbol{\theta}$. That is

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}; \mathbf{x}) &= f(\mathbf{x}; \boldsymbol{\theta}) \\ &= f(x_1 | \boldsymbol{\theta}) \times f(x_2 | x_1, \boldsymbol{\theta}) \times \dots \times f(x_n | x_1, \dots, x_{n-1}, \boldsymbol{\theta}) \\ &= f(x_1 | \boldsymbol{\theta}) \prod_{t=2}^n f(x_t | x_{t-1}, \boldsymbol{\theta}) \quad (\text{owing to the Markov property}).\end{aligned}$$

AR(1) Process: Maximum Likelihood Estimation

The required pdfs are

$$f(x_1 | \boldsymbol{\theta}) = \frac{\sqrt{1 - \phi_1^2}}{\sqrt{2\pi\sigma_z^2}} \exp \left\{ -\frac{(1 - \phi_1^2)}{2\sigma_z^2} (x_1 - \mu)^2 \right\}$$

$$f(x_t | x_{t-1}, \boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi\sigma_z^2}} \exp \left\{ -\frac{1}{2\sigma_z^2} (x_t - (\mu + \phi_1(x_{t-1} - \mu)))^2 \right\}$$

Since

$$X_1 \sim \mathcal{N} \left(\mu, \frac{\sigma_z^2}{1 - \phi_1^2} \right);$$

$$X_t | X_{t-1} = x_{t-1} \sim \mathcal{N} (\mu + \phi_1(x_{t-1} - \mu), \sigma_z^2) .$$

AR(1) Process: Maximum Likelihood Estimation

Then, the likelihood function is

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}|\mathbf{x}) &= \frac{\sqrt{1 - \phi_1^2}}{\sqrt{2\pi\sigma_z^2}} \exp \left\{ -\frac{(1 - \phi_1^2)}{2\sigma_z^2} (x_1 - \mu)^2 \right\} \\ &\times (2\pi\sigma_z^2)^{-\frac{n-1}{2}} \exp \left\{ -\frac{1}{2\sigma_z^2} \sum_{t=2}^n (x_t - (\mu + \phi_1(x_{t-1} - \mu)))^2 \right\} \\ &= (2\pi\sigma_z^2)^{-\frac{n}{2}} (1 - \phi_1^2)^{\frac{1}{2}} \exp \left\{ -\frac{1}{2\sigma_z^2} S(\mathbf{x}, \mu, \phi_1) \right\}\end{aligned}$$

where

$$\begin{aligned}S(\mathbf{x}, \mu, \phi_1) &= (1 - \phi_1^2)(x_1 - \mu)^2 + \sum_{t=2}^n (x_t - \mu - \phi_1(x_{t-1} - \mu))^2 \\ &= \sum_{t=1}^n (x_t - \mu)^2 - 2\phi_1 \sum_{t=2}^n (x_t - \mu)(x_{t-1} - \mu) \\ &\quad + \phi_1^2 \sum_{t=2}^{n-1} (x_t - \mu)^2.\end{aligned}$$

AR(1) Process: Maximum Likelihood Estimation

If we define

$$t_0 = \sum_{t=1}^n x_t, \quad t_1 = \sum_{t=2}^{n-1} x_t;$$
$$d_{00} = \sum_{t=1}^n x_t^2, \quad d_{01} = \sum_{t=2}^n x_t x_{t-1}, \quad d_{11} = \sum_{t=2}^{n-1} x_t^2;$$

then $S(\mathbf{x}, \mu, \phi_1)$ can be written as

$$S(\mathbf{x}, \mu, \phi_1) = d_{00} - 2\phi_1 d_{01} + \phi_1^2 d_{11} - 2\mu(1 - \phi_1)(t_0 - \phi_1 t_1) \\ + \mu^2(1 - \phi_1)(n - \phi_1(n - 2)).$$

AR(1) Process: Maximum Likelihood Estimation

The log-likelihood function is

$$\begin{aligned}\ell(\boldsymbol{\theta}|\mathbf{x}) &= \log \mathcal{L}(\boldsymbol{\theta}|\mathbf{x}) \\ &= -\frac{n}{2} \log(2\pi\sigma_z^2) + \frac{1}{2} \log(1 - \phi_1^2) - \frac{1}{2\sigma_z^2} S(\mathbf{x}, \mu, \phi_1)\end{aligned}$$

To find the maximum likelihood estimates $(\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2)$ of $(\mu, \phi_1, \sigma_z^2)$ we solve the score equations

$$\left. \frac{\partial \ell(\boldsymbol{\theta}|\mathbf{x})}{\partial \mu} \right|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0, \quad \left. \frac{\partial \ell(\boldsymbol{\theta}|\mathbf{x})}{\partial \phi_1} \right|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0, \quad \left. \frac{\partial \ell(\boldsymbol{\theta}|\mathbf{x})}{\partial \sigma_z^2} \right|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0.$$

AR(1) Process: Maximum Likelihood Estimation

We solve $\left. \frac{\partial \ell(\boldsymbol{\theta}|\mathbf{x})}{\partial \mu} \right|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0$

AR(1) Process: Maximum Likelihood Estimation

We solve $\left. \frac{\partial \ell(\boldsymbol{\theta}|\mathbf{x})}{\partial \phi_1} \right|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0$

AR(1) Process: Maximum Likelihood Estimation

We solve $\frac{\partial \ell(\boldsymbol{\theta}|\mathbf{x})}{\partial \sigma_z^2} \bigg|_{\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2} = 0$

AR(1) Process: Maximum Likelihood Estimation

We observe that

$$\hat{\mu} = \frac{t_0 - \hat{\phi}_1 t_1}{n - \hat{\phi}_1(n-2)};$$

$$\hat{\phi}_1 = \frac{\sum_{t=2}^n (x_t - \hat{\mu})(x_{t-1} - \hat{\mu})}{\sum_{t=2}^{n-1} (x_t - \hat{\mu})^2} - \left(\frac{\hat{\phi}_1}{1 - \hat{\phi}_1^2} \right) \frac{\hat{\sigma}_z^2}{\sum_{t=2}^{n-1} (x_t - \hat{\mu})^2};$$

$$\hat{\sigma}_z^2 = \frac{1}{n} S(\mathbf{x}, \hat{\mu}, \hat{\phi}_1).$$

Unfortunately, the above set of score equations cannot be solved analytically (i.e. by hand).

Instead, we must use a numerical method to determine the solution.

In R we would use the function `arima` (with the option `method="ML"`) to fit the model and solve the above equations.

Relationship Between Methods

When n is large, it happens that the maximum likelihood estimates $(\hat{\mu}, \hat{\phi}_1, \hat{\sigma}_z^2)$ and the method of moments estimates $(\tilde{\mu}, \tilde{\phi}_1, \tilde{\sigma}_z^2)$ are approximately the same.

This is true, in general, when estimating parameters in $AR(p)$ processes but **not** for $MA(q)$ processes.

Lecture 7: Learning Outcomes

- ▶ Know and understand the steps that should be followed for model identification when fitting an $ARMA(p, q)$ model.
- ▶ Be able to interpret various sample plots (e.g. the sample time plot, sample ACF plot - correlogram - and sample PACF plot) to aid in choosing a model to fit to a set of observed data.
- ▶ Understand that the method of moments and maximum likelihood approaches may be used for parameter estimation when fitting a model to observed time series data.
- ▶ Be able to use the method of moments approach and maximum likelihood approach to parameter estimation for the $AR(1)$ process.